

Implicit Neural Representation of Graphons for Extremal Graph Theory

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Abstract

Machine learning is opening new directions in mathematical discovery, from automatic provers for Olympiad problems to progress on long-standing conjectures. We propose a neural approach to asymptotic extremal problems in graph theory, which study the limiting behaviour of graph optimisation as the number of vertices tends to infinity. Such problems can often be reformulated as continuous optimisation over *graphons*, symmetric measurable functions $[0, 1]^2 \rightarrow [0, 1]$. We represent graphons as implicit neural networks and optimise the resulting objectives by gradient descent. Because optimal graphons are typically discontinuous (often step-like), standard neural architectures perform poorly; to address this, we use a diffusion-inspired design with multi-scale sinusoidal input encodings. Also, to enforce the constraint (e.g., fixed edge density) in such a graphon-optimisation problem, we incorporate a solver for an empirical version of the constraint into the architecture, and backpropagate through the solver via implicit differentiation. For some generalised Turán problems and domination exponent problems that previously required extensive human effort, our method rediscovers known optimal graphons without human intervention. For some open problems of the same kind, it finds novel candidate optima that may be of interest to the mathematics community.

1. Introduction

Machine learning is opening new directions in mathematical discovery, from automatic provers for Olympiad problems (Polu et al., 2023; Trinh et al., 2024; Huang & Yang, 2025), and autoformalisation tools that translate human proofs into machine-checkable form (Wu et al., 2022; Jiang

et al., 2023), to progress on long-standing conjectures (Wagner, 2021; Davies et al., 2021; He et al., 2024). In this paper, we extend this line of research by proposing a neural approach to problems in extremal graph theory.

A central object in extremal graph theory is the homomorphism density $t(H, G)$ between two finite undirected graphs H and G . It measures the probability that a uniformly random map from the vertex set $V(H)$ to the vertex set $V(G)$ is a graph homomorphism. For instance, for each n , the constrained optimisation problem

$$\max_G t(K_2, G) \text{ subject to } t(K_3, G) = 0 \wedge |V(G)| = n \quad (1)$$

is one of the most basic problems in extremal graph theory. Here K_m denotes the complete graph with m vertices, so K_2 and K_3 represent an edge and a triangle, respectively. The problem asks for a graph G with no triangles that maximises the edge density. Although the answer to this problem is well-known—a classical result of Mantel (1907)—a large number of similar optimisation problems involving homomorphism densities remain open and are actively investigated in the extremal graph theory community (Sidorenko, 1993b; Borgs et al., 2006; Razborov, 2008).

One common approach for tackling such optimisation problems is to study their asymptotic counterparts. Concretely, one considers an optimisation problem parameterised by the number of vertices n and analyses the limit of the optimal value as $n \rightarrow \infty$. Such asymptotic problems can often be reformulated as continuous optimisation problems over graphons, which are symmetric measurable functions from $[0, 1]^2$ to $[0, 1]$ and serve as limit objects of sequences of dense graphs (Lovász, 2012). By moving from discrete optimisation over finite graphs to continuous optimisation over graphons, one can compactify the search space, simplify the formulation, and bring to bear tools from continuous mathematics and optimisation¹.

Our goal is to address these optimisation problems over graphons using deep learning. We represent graphons as neural networks from $[0, 1]^2$ to $[0, 1]$, following the paradigm of implicit neural representations (Park et al., 2019; Mescheder et al., 2019; Chen & Zhang, 2019).

¹This reformulation provides a useful language, but by itself rarely resolves the underlying difficulty of these problems.

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Our approach is designed to address two recurring challenges in graphon optimisation. First, known optimal graphons are typically discontinuous (often step functions), and unknown optima are expected to have similar structure. Second, the problems often involve constraints. For example, Eq. (1) constrains the triangle density to be zero, and its graphon analogue imposes a corresponding constraint on the graphon homomorphism density. To address the first challenge, we use a diffusion-inspired neural architecture with multi-scale sinusoidal input encodings. To address the second, we incorporate an explicit constraint-solving step into the architecture and backpropagate through it via implicit differentiation, as in deep equilibrium models (Bai et al., 2019). We also introduce a training algorithm that improves sample efficiency by exploiting symmetries of the target homomorphism-density objectives.

We demonstrate the effectiveness of our architecture and training algorithm on a range of constrained and unconstrained graphon optimisation problems. We show that our method recovers known nontrivial results in extremal graph theory, including triangle-density minimisation and 5-cycle-density minimisation under an edge-density constraint (Goodman, 1959; Bollobás, 1975; Lovász & Simonovits, 1983; Fisher, 1989; Razborov, 2008), which previously required extensive human effort. We then apply our method to open problems, producing candidate minimisers for the 7-cycle density under the same edge-density constraint, and a candidate counterexample to a conjectured inequality between exponentiated homomorphism densities of the Heawood and Desargues graphs.

2. Background

For a positive integer n , let $[n] := \{1, 2, \dots, n\}$. Throughout the paper, a *graph* means an undirected finite simple graph (no self-loops and no multiple edges). For a graph G , we write $V(G)$ and $E(G)$ for its vertex and edge sets, and let $v(G) := |V(G)|$ and $e(G) := |E(G)|$. We use standard notation for familiar graphs: for example, K_n denotes the complete graph on n vertices (with vertex set $[n]$ and edge set $\{\{i, j\} : i \neq j\}$), and C_n denotes the cycle on n vertices (with vertex set $[n]$ and edge set $\{\{i, i+1\} : 1 \leq i \leq n-1\} \cup \{\{n, 1\}\}$).

Let H and G be graphs. A *homomorphism* from H to G is a map $\phi : V(H) \rightarrow V(G)$ that preserves adjacency, i.e., $\{\phi(u), \phi(v)\} \in E(G)$ for all $\{u, v\} \in E(H)$. The *homomorphism density* from H to G , denoted by $t(H, G)$, is the probability that a uniformly random map $\phi : V(H) \rightarrow V(G)$ is a homomorphism. Equivalently, one may write

$$t(H, G) := \frac{1}{v(G)^{v(H)}} \sum_{\phi: V(H) \rightarrow V(G)} \prod_{\{u, v\} \in E(H)} \mathbb{I}_{[\{\phi(u), \phi(v)\} \in E(G)]}, \quad (2)$$

where $\mathbb{I}$ is the indicator function of the corresponding event.

A large body of extremal graph theory concerns asymptotic optimisation problems of the form

$$\liminf_{n \rightarrow \infty} \left(\min_G f(t(H_1, G), \dots, t(H_k, G)) \right. \quad (3)$$

$$\left. \text{s.t. } v(G) = n \wedge \bigwedge_{j=1}^{\ell} g_j(t(F_1, G), \dots, t(F_m, G)) = 0 \right)$$

where $H_1, \dots, H_k$ and $F_1, \dots, F_m$ are graphs and $f, g_1, \dots, g_{\ell}$ are functions of reals.

We focus on two important cases of these optimisation problems. The first is to find the smallest homomorphism density of a given graph H subject to the fixed edge density p :

$$\liminf_{n \rightarrow \infty} \left(\min_G t(H, G) \text{ s.t. } v(G) = n \wedge t(K_2, G) = p \right). \quad (4)$$

This generalises Turán’s theorem (1941) and the Erdős–Stone–Simonovits theorem (1946; 1966), both classical results in extremal graph theory. In particular, for any H and $p \leq 1 - (1/(\chi(H) - 1))$, where $\chi(H)$ is the chromatic number of H , the solution to (4) is 0. If $p > 1 - (1/(\chi(H) - 1))$, the solution to (4) is known only for a handful of cases, e.g., $H = K_3$ by Razborov (2008), $H = K_4$ by Nikiforov (2011), and $H = K_r$ by Reiher (2016). Even for C_5 , the solution is known only for $p = 1 - 1/k$, for a positive integer k (Bennett et al., 2020).

The second type does not have any constraint and has the following form: for graphs H_1 and H_2 ,

$$\liminf_{n \rightarrow \infty} \left(\min_G t(H_1, G)^{m_2} - t(H_2, G)^{m_1} \text{ s.t. } v(G) = n \right). \quad (5)$$

Often, the key question is whether the asymptotic optimum in (5) is non-negative. Sidorenko’s conjecture (1993a; 1993b) predicts that this inequality holds when H_1 is bipartite, $H_2 = K_2$, and $(m_1, m_2) = (e(H_1), e(H_2))$. More systematic studies for general pairs (H_1, H_2) appear in recent work (Conlon & Lee, 2024; Behague et al., 2025).

Our results rely on an important reformulation of the asymptotic optimisation problem in Eq. (3) in terms of graphons. A *graphon* W is a symmetric measurable function from $[0, 1]^2$ to $[0, 1]$. Intuitively, it can be viewed as the adjacency matrix of a “continuum” graph with vertex set $[0, 1]$, where $W(x, y)$ specifies the edge between x and y labelled with probability. We extend homomorphism densities to graphons as follows: for a graph H with vertex set $[n]$,

$$t(H, W) := \mathbb{E}_{x_1, \dots, x_n \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]} \left[\prod_{\{i, j\} \in E(H)} W(x_i, x_j) \right]. \quad (6)$$

For continuous $f, g_1, \dots, g_{\ell}$, the asymptotic value of Eq. (3) equals the optimum of the corresponding graphon problem

$$\min_W f(t(H_1, W), \dots, t(H_k, W))$$

$$\text{s.t. } \bigwedge_{j=1}^{\ell} g_j(t(F_1, W), \dots, t(F_m, W)) = 0. \quad (7)$$

We use this graphon formulation throughout the paper.

3. Extremal Problems with Implicit Neural Representation

We tackle the graphon formulations of the optimisation problems in Eqs. (4) and (5):

$$\begin{aligned} \min_W t(H, W) \text{ s.t. } t(K_2, W) = p, \\ \min_W (t(H_1, W)^{m_2} - t(H_2, W)^{m_1}), \end{aligned} \quad (8)$$

for graphs H, H_1, H_2 , a real $p \in (0, 1)$, and positive integers m_1, m_2 , where W ranges over graphons.

Our approach represents W as a neural network $f_{\theta'}$, recasts the problems above as optimisation over the parameters θ' , and applies gradient-based training. A direct implementation of this idea fails for several reasons. First, known optimal graphons are typically step functions, which standard neural architectures such as MLPs struggle to represent due to their bias towards smooth functions. Second, each occurrence of $t(\cdot, W)$ involves intractable integrals; estimating these expectations and their powers efficiently requires careful, problem-specific sampling schemes. Finally, the constraint in the first problem is poorly handled by naïve regularisation, so the architecture must enforce it directly—e.g., by solving a sample-based version of the constraint at each forward pass. In this section we describe the neural architecture and training procedure that address these challenges.

Architecture We parameterise graphons using a residual MLP that, crucially, injects a layer-specific sinusoidal encoding of the input at every layer. When the architecture is used for constrained graphon optimisation (such as the first type in Eq. (8)), it additionally includes a constraint solver.

Concretely, we fix the number of layers $L \geq 2$, hidden width d , a differentiable activation $\phi : \mathbb{R} \rightarrow \mathbb{R}$, and an increasing function $s : [L] \rightarrow (0, \infty)$ that sets the input-encoding scale per layer; in our applications, $s(\ell) = \ell$ or $s(\ell) = 2^{\ell-1}$. The architecture defines a graphon-representing network $f_{\theta}[c]$, with a distinguished scalar parameter $c \in \mathbb{R}$ and remaining parameters θ , as follows (for $2 \leq \ell \leq L$):

$$\begin{aligned} f_{\theta}[c] : [0, 1]^2 &\rightarrow [0, 1] \\ f_{\theta}[c](x_1, x_2) &:= \text{sm}(V g_{\theta}^{(L)}(\min(x_1, x_2), \max(x_1, x_2)) + c) \\ g_{\theta}^{(1)}, g_{\theta}^{(\ell)} : \{\mathbf{x} \in [0, 1]^2 : \mathbf{x} = (x_1, x_2) \text{ and } x_1 \leq x_2\} &\rightarrow \mathbb{R}^d \\ g_{\theta}^{(1)}(\mathbf{x}) &:= \phi(W^{(1)} \sin(U^{(1)} \mathbf{x}) + b^{(1)}) \\ g_{\theta}^{(\ell)}(\mathbf{x}) &:= g_{\theta}^{(\ell-1)}(\mathbf{x}) + \\ &\quad \phi(V^{(\ell-1)} g_{\theta}^{(\ell-1)}(\mathbf{x}) + W^{(\ell)} \sin(U^{(\ell)} \mathbf{x}) + b^{(\ell)}). \end{aligned}$$

Here sm is the sigmoid function. The parameters are $c \in \mathbb{R}$, $b^{(\ell)} \in \mathbb{R}^d$, $V \in \mathbb{R}^{1 \times d}$, $U^{(\ell)} \in \mathbb{R}^{d \times 2}$, and $V^{(\ell')}, W^{(\ell)} \in \mathbb{R}^{d \times d}$, for $\ell' \in [L-1]$ and $\ell \in [L]$. They are initialised by

$$\begin{aligned} c, b_i^{(\ell)}, V_{1,i}, V_{i,j}^{(\ell')}, W_{i,j}^{(\ell)} &\stackrel{\text{iid}}{\sim} \text{Unif}[-1/\sqrt{2d}, 1/\sqrt{2d}], \\ U_{i,k}^{(\ell)} &\stackrel{\text{iid}}{\sim} \text{Unif}[-s(\ell)/\sqrt{2}, s(\ell)/\sqrt{2}], \end{aligned}$$

for all $i, j \in [d]$ and $k \in [2]$. We let θ denote all parameters except c . Note that, due to the use of $\min$ and $\max$, we have $f_{\theta}[c](x_1, x_2) = f_{\theta}[c](x_2, x_1)$ for all x_1, x_2 , so $f_{\theta}[c]$ is symmetric and thus represents a graphon.

The architecture has two notable features. First, it repeatedly injects a sinusoidal encoding of the input (x_1, x_2) into every layer, with an encoding scale that increases with depth. This design is inspired by diffusion models, where early steps capture low-frequency structure and later steps capture higher-frequency details (Salmona et al., 2022; Dieleman, 2024). We conjecture that representing step-like graphons poses a similar multi-scale modelling challenge. Increasing the scale $s(\ell)$ with depth effectively increases the frequency content of the encoding in deeper layers (Tancik et al., 2020), encouraging late layers to model sharper boundaries.

Second, in constrained settings we determine c via a constraint solver rather than learning it directly. When $f_{\theta}[c]$ is used to solve the first problem in Eq. (8), we need to satisfy the edge-density constraint $t(K_2, f_{\theta}[c]) = p$ for some $p \in (0, 1)$. At each training step (and at inference time), we assume access to a batch of K i.i.d. samples

$$\mathcal{E} = \{(x_1^{(k)}, x_2^{(k)})\}_{k \in [K]}, \quad (x_1^{(k)}, x_2^{(k)}) \stackrel{\text{iid}}{\sim} \text{Unif}([0, 1]^2).$$

Before applying $f_{\theta}[c]$ to other inputs, we solve for c using

$$\begin{aligned} \text{SOLVE}_{\theta}(\mathcal{E}, p) &:= c_0 \\ \text{such that } \frac{1}{|\mathcal{E}|} \sum_{(x_1, x_2) \in \mathcal{E}} f_{\theta}[c_0](x_1, x_2) &= p. \end{aligned} \quad (9)$$

The left-hand side of the above equation is an unbiased estimator of $t(K_2, f_{\theta}[c_0])$, so this is an empirical counterpart of the original constraint. The equation has a unique solution; the left-hand side is continuous and strictly increasing in c_0 , and it converges to 0 and 1 as $c_0 \rightarrow -\infty$ and $c_0 \rightarrow \infty$, respectively. We solve it using a variant of Newton’s method; see Algorithm 1 in the appendix. After obtaining the solution c^* , we apply $f_{\theta}[c^*]$ to other input pairs.

When $f_{\theta}[c]$ is used for the second problem in Eq. (8) (and other unconstrained graphon-optimisation problems), c is learnt by gradient-based training, along with θ .

Estimation of Homomorphism Density Let H be a graph with vertex set $[n]$. Recall that the homomorphism

density from H to a graphon $f_\theta[c]$ is: $t(H, f_\theta[c]) = \mathbb{E}_{x_1, \dots, x_n \sim \text{Unif}[0,1]} [\prod_{\{i,j\} \in E(H)} f_\theta[c](x_i, x_j)]$. A core of our training algorithm is an estimator of this quantity.

Our estimator is parameterised by a set $\mathcal{S}$ of permutations of the vertex set $[n]$. It draws (Nn) i.i.d. samples from $\text{Unif}[0, 1]$: $x_i^{(k)} \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]$ for $(k, i) \in [N] \times [n]$. Then, it computes

$$\hat{t}(H, f_\theta[c]) := \frac{1}{N|\mathcal{S}|} \sum_{k \in [N]} \prod_{\pi \in \mathcal{S}} f_\theta[c](x_{\pi(i)}^{(k)}, x_{\pi(j)}^{(k)}). \quad (10)$$

The key design choice is $\mathcal{S}$. When $\mathcal{S} = \{\text{id}\}$, $\hat{t}(H, f_\theta[c])$ reduces to the standard Monte Carlo estimator using N independent samples of $(x_1, \dots, x_n)$. When $\mathcal{S}$ contains multiple permutations and is chosen appropriately, the estimator can reuse the same N sampled tuples across the $|\mathcal{S}|$ permuted copies, often achieving substantially lower variance than using $\mathcal{S} = \{\text{id}\}$ at the same sampling cost. Also, different permutations may induce repeated evaluations of the same unordered pair $(x_i^{(k)}, x_j^{(k)})$ across summands; caching these values reduces the number of network evaluations from $(N|\mathcal{S}|e(H))$ to $(NCe(H))$ for some $C < |\mathcal{S}|$. We exploit this effect for larger graphs.

Finally, the estimator extends to unbiased estimation of polynomial expressions in homomorphism densities, such as $t(H_1, f_\theta[c])^{m_2} - t(H_2, f_\theta[c])^{m_1}$, because of the identity

$$t(H, f_\theta[c]) t(H', f_\theta[c]) = t(H \oplus H', f_\theta[c]), \quad (11)$$

where $H \oplus H'$ denotes the disjoint union of H and H' . We expand the polynomial into a linear combination of monomials, rewrite each monomial as a homomorphism density via Eq. (11), estimate each resulting density with Eq. (10), and combine the estimates linearly². We use this extension for the second problem in Eq. (8).

Training Algorithm Our training algorithm for $f_\theta[c]$ approximately solves a given graphon-optimisation problem using the estimated gradient of the optimisation objective.

Constrained Problems. Consider the first problem in Eq. (8). Fix a graph H with vertex set $[n]$, and $p \in (0, 1)$. We want to minimise $t(H, f_\theta[c])$ under the constraint $t(K_2, f_\theta[c]) = p$. At each iteration, we (approximately) enforce the constraint by solving for a scalar offset c_θ using the empirical constraint in Eq. (9), and then update θ using an estimated gradient of $t(H, f_\theta[c_\theta])$. Concretely, one update step is:

$$x_i^{(k)} \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1] \quad \text{for } (k, i) \in [N] \times [n],$$

$$\mathcal{D} := \{(x_1^{(k)}, \dots, x_n^{(k)}) : k \in [N]\},$$

²In our experiments, we used U-statistic of monomials.

Table 1. K_3 densities in graphons found by our method and optimal construction for K_3 density minimisation under $t(K_2, W) = p$.

p	Optimum	Ours
1/2	0	0.00008130
2/3	0.22222222	0.22232803 (+0.05%)
4/5	0.48	0.48010290 (+0.02%)
4/7	0.10298638	0.10299835 (+0.01%)
5/8	0.17361111	0.17362659 (+0.01%)
5/7	0.31363508	0.31370832 (+0.02%)

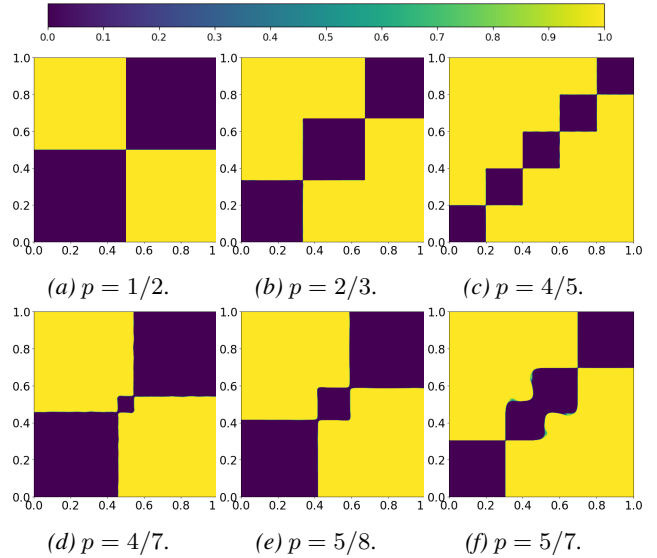


Figure 1. Graphons found by our method for K_3 density minimisation under the edge-density constraint $t(K_2, W) = p$.

$$\mathcal{E} := \{(x_i^{(k)}, x_j^{(k)}) : k \in [N], i, j \in [n], i < j\},$$

$$c_\theta := \text{SOLVE}_\theta(\mathcal{E}, p),$$

$$\theta_{t+1} \leftarrow \theta_t - \eta_t \cdot \nabla_\theta \hat{t}_\mathcal{D}(H, f_\theta[c_\theta])|_{\theta=\theta_t}.$$

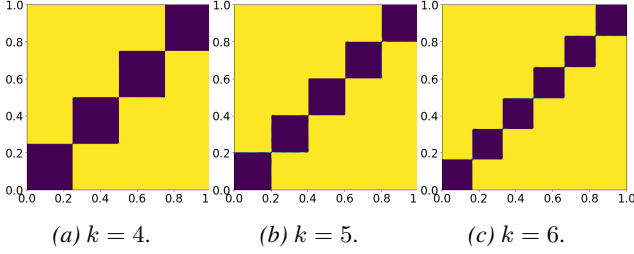
Here $\hat{t}_\mathcal{D}$ denotes the estimator of $t(H, f_\theta[c])$ in Eq. (10) evaluated on the sampled tuples in $\mathcal{D}$. We set η_t using cosine-annealing learning-rate scheduling (Loshchilov & Hutter, 2017), which helps alternate between discovering coarse structure and refining sharp boundaries. We resample $\mathcal{D}$ and $\mathcal{E}$ at every iteration (rather than reusing a fixed sample set) to mitigate overfitting to the Monte Carlo samples. Note that $\mathcal{E}$ reuses the sampled $x_i^{(k)}$ from $\mathcal{D}$, which reduces the number of forward evaluations of $f_\theta[c_\theta]$ per iteration.

By the definition of our homomorphism-density estimator, the gradient takes the form

$$\nabla_\theta \hat{t}_\mathcal{D}(H, f_\theta[c_\theta]) = \frac{1}{N|\mathcal{S}|} \sum_{\substack{k \in [N], \pi \in \mathcal{S} \\ e = \{\pi(i), \pi(j)\} \in E(H)}} \left(\left(\nabla_\theta f_\theta[c_\theta](x_{\pi(i)}^{(k)}, x_{\pi(j)}^{(k)}) \right) \prod_{\substack{\{\pi(u), \pi(v)\} \\ \in E(H) \setminus \{e\}}} f_\theta[c_\theta](x_{\pi(u)}^{(k)}, x_{\pi(v)}^{(k)}) \right).$$

Table 2. K_k densities in graphons found by our method and optimal construction for K_k density min. under $t(K_2, W) = 1 - 1/k$.

k	Optimum	Ours (Estimated)
$k = 4$	0.09375	(0.09403727, 0.09411154)
$k = 5$	0.0384	(0.03868024, 0.03872081)
$k = 6$	0.01543210	(0.01565511, 0.01568170)


 Figure 2. Graphons found by our method for K_k density minimisation under the edge-density constraint $t(K_2, W) = 1 - 1/k$.

Using implicit differentiation, we backpropagate through $c_\theta = \text{SOLVE}_\theta(\mathcal{E}, p)$ and obtain the formula for $\nabla_\theta f_\theta[c_\theta]$:

Lemma 3.1. For all $\mathbf{x} \in [0, 1]^2$,

$$\begin{aligned} \nabla_\theta f_\theta[c_\theta](\mathbf{x}) &= \frac{\partial f_\theta[c](\mathbf{x})}{\partial \theta} \Big|_{c=c_\theta} - \left(sm'(h_\theta(\mathbf{x}) + c_\theta) \right. \\ &\quad \times \sum_{\mathbf{y} \in \mathcal{E}} \frac{sm'(h_\theta(\mathbf{y}) + c_\theta)}{\sum_{\mathbf{z} \in \mathcal{E}} sm'(h_\theta(\mathbf{z}) + c_\theta)} \nabla_\theta h_\theta(\mathbf{y}) \Big), \end{aligned}$$

where $h_\theta(x_1, x_2) := Vg_\theta^{(L)}(\min(x_1, x_2), \max(x_1, x_2))$.

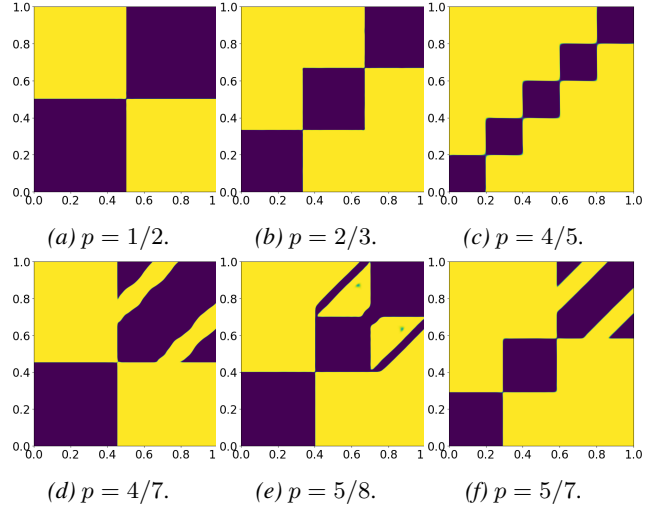
Unconstrained Problems. Now consider the second problem in Eq. (8). Let H_1 and H_2 be graphs with vertex sets $[n_1]$ and $[n_2]$. Fix positive integers m_1, m_2 . We minimise $t(H_1, f_\theta[c])^{m_2} - t(H_2, f_\theta[c])^{m_1}$. Using Eq. (11), we rewrite this as $t(H_1^{\oplus m_2}, f_\theta[c]) - t(H_2^{\oplus m_1}, f_\theta[c])$, where $H^{\oplus m}$ denotes the disjoint union of m copies of H . We then optimise both θ and c by gradient descent. One update step is:

$$\begin{aligned} x_i^{(k)} &\stackrel{\text{iid}}{\sim} \text{Unif}[0, 1] \quad \text{for } (k, i) \in [N] \times [m_2 n_1], \\ \mathcal{D} &:= \{(x_1^{(k)}, \dots, x_{m_2 n_1}^{(k)}) : k \in [N]\}, \\ y_j^{(k)} &\stackrel{\text{iid}}{\sim} \text{Unif}[0, 1] \quad \text{for } (k, j) \in [N] \times [m_1 n_2], \\ \mathcal{D}' &:= \{(y_1^{(k)}, \dots, y_{m_1 n_2}^{(k)}) : k \in [N]\}, \\ (\theta_{t+1}, c_{t+1}) &\leftarrow (\theta_t, c_t) - \eta_t \times \left(\nabla_{(\theta, c)} \left(\hat{t}_{\mathcal{D}}(H_1^{\oplus m_2}, f_\theta[c]) - \hat{t}_{\mathcal{D}'}(H_2^{\oplus m_1}, f_\theta[c]) \right) \Big|_{(\theta, c) = (\theta_t, c_t)} \right). \end{aligned}$$

We sample $\mathcal{D}, \mathcal{D}'$ fresh at each step, instead of sampling them in the first step and reusing them subsequently. This is to prevent the network from overfitting particular datasets.

 Table 3. C_5 densities in graphons found by our method and conjectured construction for C_5 density min. under $t(K_2, W) = p$.

p	Conjectured	Ours
1/2	0	0.00004086
2/3	0.12345679	0.12350364 (+0.04%)
4/5	0.3264	0.32644549 (+0.01%)
4/7	0.04587149	0.04590020 (+0.06%)
5/8	0.08653616	0.08657084 (+0.04%)
5/7	0.18200809	0.18206173 (+0.03%)


 Figure 3. Graphons found by our method for C_5 density minimisation under the edge-density constraint $t(K_2, W) = p$.

4. Experiment Results

We empirically evaluate our method on both resolved and unresolved problems in extremal graph theory that fit one of the forms in Eq. (8). The goals are (i) to test whether the method can rediscover known optima that required substantial human effort, and (ii) to investigate whether it can produce mathematically interesting candidates for open problems. We also include an ablation study that assesses the impact of key components of our architecture and training procedure.

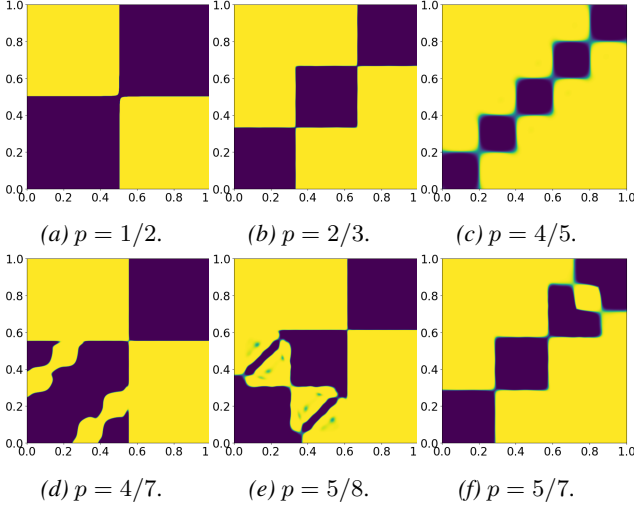
In this section, we visualise graphons as heatmaps using a blue-to-yellow colormap (Viridis), where blue represents 0 and yellow represents 1; see Fig. 1. Reported homomorphism densities are either numerically computed by discretising the learnt graphon into an $M \times M$ weighted adjacency matrix, or estimated using 2^{28} samples with a 95% bootstrap confidence interval (shown in *italic font*) when discretisation is computationally infeasible or memory-intensive. Across multiple training runs, we report the run with the lowest objective value. See Appendix C for details.

Constrained Homomorphism-Density Minimisation

We apply our method to the problem of minimising the homomorphism density of K_n under an edge-density con-

Table 4. C_7 densities in graphons found by our method and conjectured construction for C_7 density min. under $t(K_2, W) = p$.

p	Conjectured	Ours
1/2	0	0.00001517
2/3	0.05763169	0.05763195 (+0.00%)
4/5	0.209664	0.20967621 (+0.01%)
4/7	0.01713510	0.01715017 (+0.09%)
5/8	0.03610272	0.03611957 (+0.05%)
5/7	0.09455118	0.09457182 (+0.02%)


 Figure 4. Graphons found by our method for C_7 density minimisation under the edge-density constraint $t(K_2, W) = p$.

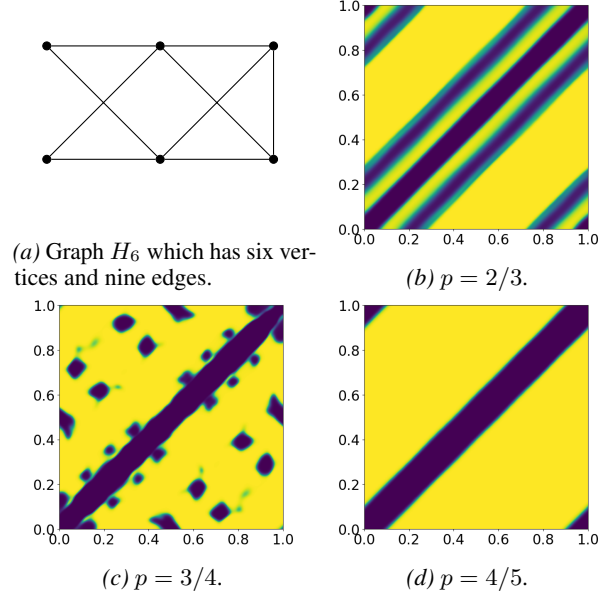
straint $t(K_2, W) = p$. This is an instance of the first problem in Eq. (8), with $H = K_n$. For $n = 3$ (triangles), this optimisation problem has been studied extensively (Goodman, 1959; Bollobás, 1975; Lovász & Simonovits, 1983; Fisher, 1989; Razborov, 2008), and its full solution was established by Razborov (2008). For general n , the full solution was obtained more recently by Reiher (2016).

We begin with the case $n = 3$. When $p = 1 - 1/k$ for some integer $k \geq 2$, the optimal graphon is the complete k -partite graphon or any of its measure-preserving permutations. For $i \in [k]$, let $X_i = [(i-1)/k, i/k]$. The complete k -partite graphon W_k is defined as follows: $W_k(x_1, x_2) = 1$ if $x_1 \in X_i$ and $x_2 \in X_j$ with $i \neq j$; otherwise, $W_k(x_1, x_2) = 0$. For a measure-preserving ϕ on $[0, 1]$, its permutation is $W_{k,\phi}(x_1, x_2) = W_k(\phi(x_1), \phi(x_2))$. Every optimum is of this form. When $1 - 1/k < p < 1 - 1/(k+1)$, the minimisation admits multiple types of optima. One example is a variant W_p of W_{k+1} , where the first k parts are equal in length but the last part X_{k+1} is shorter. Again, its permutations $W_{p,\phi}$ by measure-preserving ϕ on $[0, 1]$ are also optimal, though W_p and $W_{\phi,p}$ do not exhaust all optima.

We considered $p \in \{1/2, 2/3, 4/5, 4/7, 5/8, 5/7\}$, and trained our network for all the six cases of p . In the first

 Table 5. H_6 densities in graphons found by our method and standard construction for H_6 density minimisation under $t(K_2, W) = p$.

p	Ours	k-partite	Constant
2/3	0.02476414	0.03292181	0.02601229
3/4	0.07218717	0.07617188	0.07508469
4/5	0.12985603	0.13056	0.13421773


 Figure 5. Graphons found by our method for H_6 density minimisation under the edge-density constraint $t(K_2, W) = p$.

three cases (i.e., $p \in \{1/2, 2/3, 4/5\}$), our method produced near-optimal graphons (top row of Fig. 1) whose estimated objectives (via discretisation) are within 0.1% of the theoretical optima. In those cases, $p = 1 - 1/k$ for some integer k , and the learnt graphons closely match the complete k -partite graphons W_k described above. While measure-preserving permutations yield graphons with equivalent objective values but more complex discontinuity patterns, our method consistently converged to the simpler representatives shown in Fig. 1. For $p \in \{4/7, 5/8\}$, which satisfy $1 - 1/k < p < 1 - 1/(k+1)$ for the corresponding k , the learnt graphons (Figs. 1d and 1e) capture the core structure of the known optimal W_p described above. For $p = 5/7$, the learnt graphon (Fig. 1f) does not match the optimal structure W_p described, but the bipartite middle block structure matches the characterisation of all optimal graphons described in (Pikhurko & Razborov, 2017).

For K_k with $k > 3$, we applied our method to the following optimisation problems: $\min_W t(K_k, W)$ subject to $t(K_2, W) = 1 - 1/k$ for $k \in \{4, 5, 6\}$. The complete k -partite graphons and their permutations achieve optimal values of these problems. Despite the increasing complexity of these problems in the structures of their optimal graphons

Table 6. Difference of normalised homomorphism densities of K_r, K_s in graphons, found by our method.

	$t(K_r, W)^{(s)} - t(K_s, W)^{(r)}$
$r = 3, s = 4$	-0.05948110
$r = 3, s = 5$	-0.24935071
$r = 4, s = 5$	-0.07295104

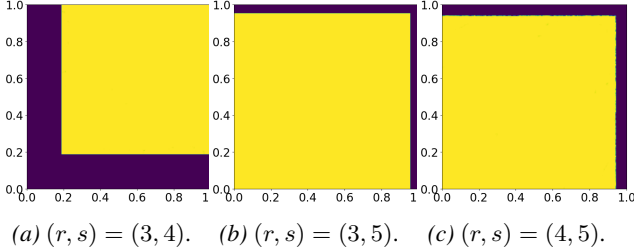


Figure 6. Graphons found by our method for the minimisation of $t(K_r, W)^{(s)} - t(K_s, W)^{(r)}$.

and the sample and computational complexities of our estimator, our method found near-optimal graphons in Fig. 2, which show the desired complete k -partite structure.

Unresolved Problems: C_5 and C_7 . We next apply our method to three graphon optimisation problems that, to the best of our knowledge, remain open: minimising the homomorphism densities of C_5, C_7 , each under a fixed edge density p . In Appendix E, we also report our results on applying our method to the Petersen graph, for which no prior results are known due to its large size and complexity.

For C_5 , partial results are known. When $p = 1 - 1/k$ for some k , Bennett et al. (2020) characterised the minimisers; for other values of p , they conjectured both the minimisers and the optimum values. We applied our method for $p \in \{1/2, 2/3, 4/5, 4/7, 5/8, 5/7\}$, which lie in the conjectured regime. The learnt graphons are shown in Fig. 3. As shown in Tab. 3, the resulting values agree with the benchmark optima (proved for $p \in \{1/2, 2/3, 4/5\}$ and conjectured for the remaining p values) within 0.1% relative error, providing empirical support for the conjectured regime. Notably, when $p = 1 - 1/k$ the learnt graphon recovers the complete k -partite structure (top row), whereas for other values of p it exhibits pronounced diagonal (banded) structure (bottom row), suggesting a qualitatively different extremal configuration away from the rational breakpoints.

For C_7 , to the best of our knowledge there are no established results. Motivated by the C_5 results of Bennett et al. (2020), we propose a conjecture for the structure and optimum value of C_7 -minimising graphons under a fixed edge density; see Appendix D. Empirically, the C_7 densities achieved by our learnt graphons match the conjectured optima with relative errors below 0.1% across all tested values of p (Tab. 4). For $p \in \{1/2, 2/3, 4/5\}$ (i.e., $p = 1 - 1/k$), the learnt graphons

Table 7. K_3 densities in graphons found by our method and ablations for K_3 density minimisation under $t(K_2, W) = 7/9$.

	$t(K_3, W)$
Ours	0.427215
ResNet	0.428968
Constant-scale s	0.428190
Regulariser	0.434445
Constant LR (low)	0.430071
Constant LR (high)	0.431645

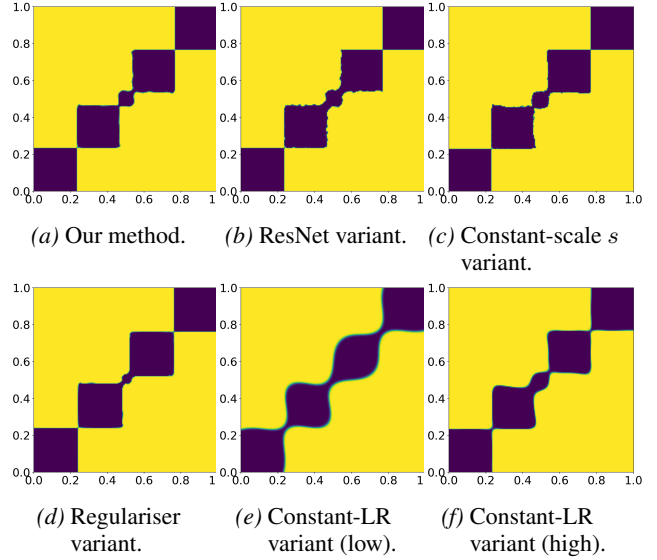


Figure 7. Graphons found by our method and its ablations, for K_3 density minimisation under the constraint $t(K_2, W) = 7/9$.

recover the complete k -partite structure, up to small numerical artefacts (top row of Fig. 4), consistent with the behaviour observed for triangles and C_5 . For $p \in \{4/7, 5/8\}$, the learnt graphons show similar patterns to those found for C_5 , suggesting that this diagonal structure is a common feature of odd-cycle minimisation. For $p = 5/7$, however, the learnt graphon exhibits a qualitatively different structure, with block-like patterns rather than the diagonal pattern seen for C_5 at the same edge density (bottom-right of Fig. 4).

Exhaustive Study over Small Graphs. We now describe a systematic study enabled by our method. We applied our method to the first problem in Eq. (8) for non-bipartite graphs H with $v(H) \leq 6$ and $e(H) \leq 9$, and with no isolated vertices, for a total of 175 graphs. For each H , we trained our network to minimise $t(H, W)$ subject to $t(K_2, W) = p$, for $p \in \{\frac{k-2}{k-1}, \frac{3k-5}{3k-2}, \frac{2k-3}{2k-1}, \frac{3k-4}{3k-1}, \frac{k-1}{k}\}$, where k is the chromatic number of H . These values of p are chosen to probe the structure of the curve $x \mapsto \min_{t(K_2, W)=x} t(H, W)$: the first and last values correspond to breakpoints of piecewise-defined curves established for several families of graphs (Razborov, 2008; Reiher, 2016; Bennett et al., 2020), while the remaining values lie in

Table 8. Homomorphism densities of Heawood (H) and Desargues (D) in graphon found by our method, and their normalised difference.

Density	Estimated Density	95% Confidence Interval
$t(H, W)$	6.79×10^{-25}	$(6.45, 7.18) \times 10^{-25}$
$t(D, W) \frac{v(H)}{v(D)}$	1.52×10^{-24}	$(5.32, 29.1) \times 10^{-25}$
Difference	Mean Diff	Conservative Diff ³
$t(H, W) - t(D, W) \frac{v(H)}{v(D)}$	-8.40×10^{-25}	1.86×10^{-25}

³Calculated as $(\text{Upper CI}_A - \text{Lower CI}_B)$. A negative value confirms $A < B$ with high confidence, so our value indicates statistical uncertainty about our candidate counterexample.

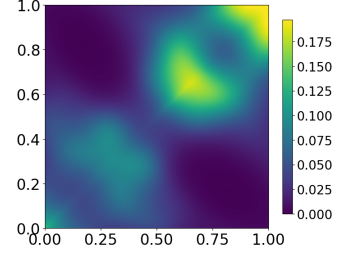


Figure 8. Graphon found for the min. task induced by the conjectured Heawood vs. Desargues.

the interior of the corresponding segments. We also implemented an automated tool for computing lower bounds on the minimum H -density using flag algebras (Razborov, 2007), which we used in our experiments.

Across 76.5% of the (H, p) instances, our method produced a graphon whose H -density closely matches the corresponding flag-algebra lower bound, suggesting that the solutions are optimal or near-optimal. Among the remaining instance, 11.8% were clearly suboptimal or failed to train, while for the remaining 11.7% we obtained graphons that appear optimal, but for which our flag-algebra bounds were not tight.

We analysed the curve $x \mapsto \min_{t(K_2, W)=x} t(H, W)$, for each H , by interpolating between the obtained points and examining whether it appears piecewise concave (as is the case for cliques (Reiher, 2016)) or convex (as conjectured for odd cycles (Bennett et al., 2020)). We found that apparent piecewise concavity is associated with graphs that are close to complete graphs, in the sense that they become complete after removing 1 or 2 edges and isolated vertices. For the other graphs, the interpolated curves appeared convex.

Finally, our experiments suggest a novel family of extremal graphon structures. As seen in Figs. 1, 3 and 4, when $p = 1 - 1/k$ for an integer $k \geq 2$, the minimisers for several problems are uniquely characterised (up to measure-preserving transformations) as complete k -partite graphons. However, our results indicate that this phenomenon is not universal: for the graph H_6 visualised in Fig. 5a at $p = 1 - 1/k$, our method finds a candidate minimiser that is neither complete k -partite nor a constant graphon (Fig. 5). These candidate graphons exhibit a geometric pattern: when the interval $[0, 1]$ is interpreted as a circle, edge probabilities appear to depend primarily on the circular distance.

Normalised Inequalities between Densities In our experiments with the problem of the second type in Eq. (8), our goal was to *disprove*, for some problem instances (H_1, H_2, m_1, m_2) , that the minimum values of the instances are non-negative, equivalently, the inequality $t(H_1, W)^{m_2} \geq t(H_2, W)^{m_1}$ holds for all graphons W .

Our first task was to disprove, for each $r, s \in \{3, 4, 5\}$ with $r < s$, that the inequality $t(K_r, W)^{\binom{s}{2}} \geq t(K_s, W)^{\binom{r}{2}}$ holds for all W . We note that such an inequality holds always if and only if $(\min_W t(K_r, W)^{\binom{s}{2}} - t(K_s, W)^{\binom{r}{2}}) \geq 0$. We applied our method to find a graphon witnessing that the minimum value here is negative. For all cases of r and s , our method found graphons (Fig. 6) that strongly suggest the failure of the inequalities; the optimisation objectives at these graphons were estimated to be negative. See Tab. 6. Our next task targets an open conjecture of Perarnau & Perkins (2018), which posits, for any graphon W and any 3-regular graph G with girth at least 6, that $t(\text{Heawood}, W) \geq t(G, W)^{v(\text{Heawood})/v(G)}$ holds for all W . We consider the Desargues graph D as a potential counterexample. Using our method, we found a candidate counterexample W described in Tab. 8 and Fig. 8.

Ablation Study We perform an ablation study on triangle-density minimisation under the constraint $t(K_2, W) = 7/9$, which is not of the form $1 - 1/k$ and admits multiple types of optima. We consider four variants: (1) replacing our architecture with a ResNet that uses sinusoidal input encoding only in the first layer; (2) using a constant per-layer input encoding scale s ; (3) enforcing the constraint via a regulariser; and (4) using a constant learning rate instead of setting it according to the cosine-annealing scheduler. For fair comparison, we performed hyperparameter searches for the constant value of s in (2), the strength of the regulariser in (3), and the constant learning rate in (4). The graphons found by these variants are shown in Fig. 7. The ResNet variant (1) and the constant-scale variant (2) often fail to recover sharp, straight boundaries, consistent with a stronger bias toward smooth functions. With the penalty-based constraint (regulariser) (3), the learnt graphon often violates fine structural details (e.g., the size of the smallest block). Without cosine-annealing scheduling (4), the learnt graphon either fails to capture the multi-block structure (for low learning rates) or exhibits less stable/sharper boundaries (for high learning rates), highlighting the benefit of alternating between higher and lower learning rates. Quantitative results after discretisation are reported in Tab. 7.

Impact Statement

This paper presents an implicit neural representation based framework to solve extremal graph theory problems, with the goal of advancing extremal graph theory by enabling more automatic and exhaustive exploration. By enabling automatic conjecture generation and testing, our method has the potential to accelerate discoveries in combinatorics and related fields. There are no major foreseeable negative societal impacts of this research.

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A. Related Work

Razborov’s *flag algebra* framework (Razborov, 2007) enables the proofs of inequalities between graph homomorphism densities by means of sums of squares (ultimately, applications of the Cauchy–Schwarz inequality) in an algebra of graphs, where each finite graph can be interpreted as a functional on the space of graphons. This viewpoint enables the automation of Cauchy–Schwarz-style arguments via semidefinite programming (SDP), and has been implemented in the package FLAGMATIC (Falgas-Ravry & Vaughan, 2013). Flag algebras and Flagmatic have resolved many open problems in extremal combinatorics over the past decade (for a survey, see Razborov (2013)), but their computational cost can be prohibitive³. This limits applicability to some instances we consider, including problems involving the Petersen, Heawood, or Desargues graph.

For fixed values of n (i.e., without taking the limit in Eqs. (4) and (5) or related formulations), one may instead attempt to solve the discrete optimisation problem directly. A classical approach is local (e.g., Tabu) search, which remains useful in combination with reinforcement learning; see, e.g., Parczyk et al. (2024); Mehrabian et al. (2024). As a direct reinforcement-learning approach, Mehrabian et al. (2024) used ALPHAZERO to find the maximum number of edges in an n -vertex graph with no copies of K_3 and C_4 . Recently, the so-called *deep cross entropy method* built upon reinforcement learning was used by Wagner (2021) for analysing discrete problems in combinatorics. It iteratively trains a neural network to learn a policy that promotes the generation of high quality examples, which succeeded in disproving two conjectures regarding the eigenvalue of the adjacency matrix and matching number or proximity. For discrete objectives, these approaches can be more versatile than our method because they can optimise non-differentiable quantities (e.g., the matching number). In contrast, they operate at a fixed graph size, whereas our method targets asymptotic problems in the dense-graph limit. Finally, PATTERNBOOST (Charton et al., 2024) takes a hybrid approach that combines local search with reinforcement learning; one illustrative example is generating lower-bound examples related to Mantel’s theorem by solving finite- n instances of Eq. (4) for $H = K_3$ (for small n) up to edge density $p \leq 1/2$. Because many combinatorial objects are highly symmetric, solutions produced by such discrete approaches can be difficult to interpret or distil into human-understandable structure. In contrast, as demonstrated by visualisations of learned graphons, our method often captures structure that is amenable to inspection and interpretation.

From the perspective of implicit neural representations, the work most closely related to ours is Munding et al. (2025), which applies implicit neural representations to the Hadwiger–Nelson problem: colouring the plane $\mathbb{R}^2$ with the minimum number of colours such that no two points at unit distance share a colour. Analogously to our approach, they represent a colouring by a neural network and recover a 6-colouring by minimising a monochromatic edge-density objective. With human post-processing, they construct two new 6-colourings that expand the known range of realisable distances from $[0.415, 0.447]$ to $[0.354, 0.657]$. In contrast, we focus on asymptotic extremal problems in graph theory that can be phrased as constrained optimisation over graphons. Moreover, our approach integrates problem structure into the learning procedure, including an architectural bias toward discontinuous, step-like solutions, explicit constraint enforcement via an embedded solver, and variance reduction in homomorphism density estimation, with the aim of supporting systematic conjecture generation and exploration across diverse extremal problems in graph theory.

B. Unbiasedness of Our Density Estimator and Proof of Lemma 3.1

Lemma B.1. *The estimator in Eq. (10) is unbiased, i.e., $\mathbb{E}[\hat{t}(H, f_\theta[c])] = t(H, f_\theta[c])$.*

³For example, to derive a nontrivial inequality for a graph G , FLAGMATIC typically enumerates all graphs H with $v(H) \geq v(G) + 1$ and solves an SDP over vectors indexed by such graphs. Graphs H with more than eight vertices are currently unsupported due to computational complexity.

Proof. We compute the claimed equality as follows:

$$\begin{aligned}
 \mathbb{E} [\widehat{t}(H, f_\theta[c])] &= \mathbb{E} \left[\frac{1}{N|\mathcal{S}|} \sum_{k \in [N], \pi \in \mathcal{S}} \left(\prod_{\{\pi(i), \pi(j)\} \in E(H)} f_\theta[c](x_{\pi(i)}^{(k)}, x_{\pi(j)}^{(k)}) \right) \right] \\
 &= \frac{1}{N|\mathcal{S}|} \sum_{k \in [N], \pi \in \mathcal{S}} \mathbb{E} \left[\prod_{\{\pi(i), \pi(j)\} \in E(H)} f_\theta[c](x_{\pi(i)}^{(k)}, x_{\pi(j)}^{(k)}) \right] \\
 &= \frac{1}{N|\mathcal{S}|} \sum_{k \in [N], \pi \in \mathcal{S}} \mathbb{E}_{x_1, \dots, x_n \stackrel{\text{iid}}{\sim} \text{Unif}[0,1]} \left[\prod_{\{i,j\} \in E(H)} f_\theta[c](x_i, x_j) \right] \\
 &= \frac{1}{N|\mathcal{S}|} \sum_{k \in [N], \pi \in \mathcal{S}} t(H, f_\theta[c]) = t(H, f_\theta[c]).
 \end{aligned}$$

□

Proof of Lemma 3.1. Let $\mathbf{x} \in [0, 1]^2$. Note that

$$\begin{aligned}
 \nabla_\theta f_\theta[c_\theta](\mathbf{x}) &= \frac{\partial f_\theta[c](\mathbf{x})}{\partial \theta} \Big|_{c=c_\theta} + \frac{\partial f_\theta[c](\mathbf{x})}{\partial c} \Big|_{c=c_\theta} \nabla_\theta c_\theta \\
 &= \frac{\partial f_\theta[c](\mathbf{x})}{\partial \theta} \Big|_{c=c_\theta} + sm'(h_\theta(\mathbf{x}) + c_\theta) \nabla_\theta c_\theta
 \end{aligned}$$

Thus, it suffices to show that

$$\nabla_\theta c_\theta = - \sum_{\mathbf{y} \in \mathcal{E}} \frac{sm'(h_\theta(\mathbf{y}) + c_\theta)}{\sum_{\mathbf{z} \in \mathcal{E}} sm'(h_\theta(\mathbf{z}) + c_\theta)} \nabla_\theta h_\theta(\mathbf{y}).$$

We prove the above equality by applying implicit differentiation to the following equation about c_θ , which holds by the choice of c_θ :

$$\frac{1}{|\mathcal{E}|} \sum_{\mathbf{y} \in \mathcal{E}} f_\theta[c_\theta](\mathbf{y}) = p.$$

First, we multiply both sides of the equation by $|\mathcal{E}|$, and differentiate them with respect to θ :

$$\left(\sum_{\mathbf{y} \in \mathcal{E}} \frac{\partial f_\theta[c](\mathbf{y})}{\partial \theta} \Big|_{c=c_\theta} \right) + \left(\sum_{\mathbf{z} \in \mathcal{E}} \frac{\partial f_\theta[c](\mathbf{z})}{\partial c} \Big|_{c=c_\theta} \nabla_\theta c_\theta \right) = 0$$

Thus,

$$\begin{aligned}
 \nabla_\theta c_\theta &= - \left(\sum_{\mathbf{y} \in \mathcal{E}} \frac{\partial f_\theta[c](\mathbf{y})}{\partial \theta} \Big|_{c=c_\theta} \right) / \left(\sum_{\mathbf{z} \in \mathcal{E}} \frac{\partial f_\theta[c](\mathbf{z})}{\partial c} \Big|_{c=c_\theta} \right) \\
 &= - \sum_{\mathbf{y} \in \mathcal{E}} \frac{sm'(h_\theta(\mathbf{y}) + c_\theta)}{\sum_{\mathbf{z} \in \mathcal{E}} sm'(h_\theta(\mathbf{z}) + c_\theta)} \nabla_\theta h_\theta(\mathbf{y}),
 \end{aligned}$$

as desired.

□

C. Details of Experiments

All experiments were conducted on a single NVIDIA RTX A5000 GPU. The execution time depends on the size of the graph, from 30 minutes for a triangle to 3 hours for the Desargues graph.

In all experiments, we used a network of width $d = 256$ and depth $L = 6$, which was sufficient to represent sharp boundaries and structural properties. The activation function ϕ was either sin or the GELU nonlinearity; we did not observe a consistent advantage of one over the other.

The set $\mathcal{S}$ in Equation Eq. (10) was chosen according to the graph. For example, for complete graphs we used $\mathcal{S} = \{\text{id}\}$, since any permutation yields the same summand. For cycles, we used a set of representatives for the cosets $S_{V(C_k)}/\text{Aut}(C_k)$, where $\text{Aut}(C_k)$ is the automorphism group of C_k , which gives $|\mathcal{S}| = (k-1)!/2$. For the other graphs, we used the set $\mathcal{S} = \{\pi^k : k \in \mathbb{N}\}$ generated by some permutation $\pi \in S_{V(G)}$, where π is chosen so that the set

$$\{\{\pi^k(v), \pi^k(w)\} : \{v, w\} \in E(G) \text{ and } k \in \mathbb{N}\}$$

equals the set of all unordered pairs of distinct vertices $\{\{v, w\} : v, w \in V(G) \text{ and } v \neq w\}$.

For training, we used the largest number of samples that fit in GPU memory, ranging from 2^{12} to 2^{16} . For large graphs such as the Petersen graph, we repeated multiple forward passes to accumulate gradients, yielding an effective sample size of 2^{16} .

For optimisation, we used the Adam optimiser with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. We repeated the experiments with learning rates $\eta \in \{10^{-3}, 5 \times 10^{-4}, 10^{-4}, 5 \times 10^{-5}\}$, and set the minimum learning rate in the scheduler to 1% of the initial learning rate. We trained for 20,000 epochs, corresponding to five cycles of cosine-annealed learning-rate scheduling.

For the constraint solver, we found that applying Newton’s method directly (without modification) can be unstable, owing to vanishing derivatives in many regimes. In particular, since many optimal graphons take values close to 0 or 1, the sigmoid derivative $sm(x)(1 - sm(x))$ is close to zero over much of the domain. Consequently, the denominator in the Newton update can become very small, leading to excessively large updates and unstable convergence. To address this, we use a clipped variant of Newton’s method. To set the clipping range, we compute upper and lower bounds on the solution c^* and use their difference as the initial clipping range. We then gradually decrease the clipping range as the iteration proceeds, which improves stability. The detailed algorithm is described in Algorithm 1.

Algorithm 1 Edge-Density Constraint Solver

Require: $p \in (0, 1), y \in \mathbb{R}^N$

Ensure: $\frac{1}{N} \sum_{i=1}^N sm(y_i + c^*) \cong p$

$c_1 \leftarrow sm^{-1}(p) - \frac{1}{N} \sum_{i=1}^N y_i$

{Initialise with guess}

$D = \max_i \{y_i\} - \min_i \{y_i\}$

{Compute difference between upper bound and lower bound}

for $t = 1, 2, \dots$ **do**

$f(c_t) \leftarrow \frac{1}{N} \sum_{i=1}^N sm(y_i + c_t) - p$

if $|f(c_t)| < \epsilon$ **then**

break

end if

$f'(c_t) = \frac{1}{N} \sum_{i=1}^N sm(y_i + c_t)(1 - sm(y_i + c_t))$

$c_{t+1} \leftarrow c_t - \text{Clip}_{[-D/t, D/t]} \left(\frac{f(c_t)}{f'(c_t)} \right)$

{Newton’s method step with bound range as clipping range}

end for

Return c_t

For the variants in the ablation study, we performed hyperparameter searches for each variant. When using constant per-layer input encoding scale s , we searched over $s(l) \in \{1, 2, 4, 8, 16\}$. For the strength of the regulariser, we searched over $\lambda \in \{0.01, 0.1, 1, 10, 100\}$. For the constant learning rate, we searched over the same set of values as the initial learning rates in the main experiments.

When reporting the homomorphism density computed by discretisation, we use $(2^{14} \times 2^{14})$ grid for K_3, C_5, C_7 and $(2^{10} \times 2^{10})$ grid for H_6 , and 64-bit floating point precision.

The environment setup and implementation to replicate our experiments can be seen in the following anonymous repository. <https://anonymous.4open.science/r/INR4AsymptoticExtremalGraphTheory-47DB/README.md>.

D. Conjecture on Odd Cycles

In this section, we provide further details of our conjecture on odd-cycle density minimisation.

In Bennett et al. (2020), the optimal graphon’s structure is constructed as follows. Let k be the integer satisfying $1 - \frac{1}{k} < p < 1 - \frac{1}{k+1}$. First, we partition the interval $[0, 1]$ into $(k+1)$ sets, $X_1, \dots, X_{k-1}, Y_1, Y_2$. We set the lengths of $X_1, \dots, X_{k-1}$

to be x , and the lengths of Y_1 and Y_2 to be $\frac{y}{2}$, where $y = 1 - (k-1)x$. We then define the graphon by

$$W(x_1, x_2) = \begin{cases} 1 & \text{if } x_1 \in X_m, x_2 \in X_n, m \neq n \\ 1 & \text{if } x_1 \in X_m, x_2 \in Y_n, \text{ or vice versa} \\ 0 & \text{if } x_1 \in X_m, x_2 \in X_m, \\ \rho & \text{if } x_1 \in Y_1, x_2 \in Y_2 \text{ or vice versa} \\ 0 & \text{if } x_1 \in Y_m, x_2 \in Y_m, \end{cases}$$

where ρ is chosen to satisfy the edge-density constraint.

One can compute the C_5 density of this graphon, which can be written as a polynomial $f_5(x, y, \rho)$ given by

$$\begin{aligned} f_5(x, y, \rho) &= \left(\frac{(k-1)_5}{10} + \frac{(k-1)_4}{2} + \frac{(k-1)_3}{2} \right) x^5 \\ &+ \left(\frac{(k-1)_4}{2} + \frac{3(k-1)_3}{2} + \frac{(k-1)_2}{2} \right) x^4 y \\ &+ \left(\left(\frac{1}{2} + \frac{\rho}{2} \right) (k-1)_3 + \left(1 + \frac{\rho}{2} \right) (k-1)_2 \right) x^3 y^2 \\ &+ \left(\left(\frac{\rho}{2} + \frac{\rho^2}{2} \right) (k-1)_2 + \frac{\rho}{2} (k-1) \right) x^2 y^3 \\ &+ \frac{\rho^3}{2} (k-1) x y^4, \end{aligned}$$

where $(k-1)_r$ is a falling factorial, $(k-1)_r = (k-1) \cdot (k-2) \cdots (k-r)$. With these definitions, the conjectured optimal value is obtained by optimising over x , y , and ρ . Equivalently, one can compute the conjectured optimum by solving

$$\begin{aligned} &\min f_5(x, y, \rho) \\ &\text{subject to } (k-1)x + y = 1, \\ &\quad (k-1)_2 x^2 + 2(k-1)xy + \rho y^2 = p, \\ &\quad x, y \geq 0, \quad 0 \leq \rho \leq 0.5. \end{aligned}$$

Using the same structure, we can generalise the conjecture to arbitrary odd cycles $C_{2\ell+1}$ by replacing f_5 with the corresponding $f_{2\ell+1}$. For instance, f_7 is given as follows.

$$\begin{aligned} f_7(x, y, \rho) &= \left(\frac{(k-1)_7}{14} + (k-1)_6 + 4(k-1)_5 + 5(k-1)_4 + \frac{3(k-1)_3}{2} \right) x^7 \\ &+ \left(\frac{(k-1)_6}{2} + 5(k-1)_5 + \frac{25(k-1)_4}{2} + \frac{15(k-1)_3}{2} + \frac{(k-1)_2}{2} \right) x^6 y \\ &+ \left((k-1)_5 + (3\rho + 7)(k-1)_4 + \left(\frac{7\rho}{2} + 10 \right) (k-1)_3 + \left(\frac{\rho}{2} + 2 \right) (k-1)_2 \right) x^5 y^2 \\ &+ \left(\left(\frac{3\rho}{2} + \frac{1}{2} \right) (k-1)_4 + \left(\frac{3\rho^2}{2} + 6\rho + \frac{5}{2} \right) (k-1)_3 + \left(\frac{\rho^2}{2} + 3\rho + 2 \right) (k-1)_2 \right) x^4 y^3 \\ &+ \left(\left(\frac{\rho^3}{2} + \frac{3\rho^2}{2} + \frac{\rho}{2} \right) (k-1)_3 + \left(\frac{\rho^3}{2} + 3\rho^2 + \frac{3\rho}{2} \right) (k-1)_2 + \frac{\rho}{2} (k-1) \right) x^3 y^4 \\ &+ \left(\left(\frac{\rho^4}{2} + \rho^3 \right) (k-1)_2 + \rho^3 (k-1) \right) x^2 y^5 \\ &+ \frac{\rho^5}{2} (k-1) x y^6. \end{aligned}$$

Table 9. Homomorphism densities of the Petersen graph in graphons under the edge-density constraint $t(K_2, W) = p$, which are found by our method and hand-crafted construction.

p	Ours (Estimated)	Hand-crafted
1/2	(0.00000000, 0.00000000)	0
2/3	(0.00184293, 0.00184619)	0.0020322
4/5	(0.03412074, 0.03413641)	0.0340869
4/7	(0.00006195, 0.00006209)	0.0001736
5/8	(0.00054270, 0.00054310)	0.0007531
5/7	(0.00571939, 0.00572465)	0.0059452

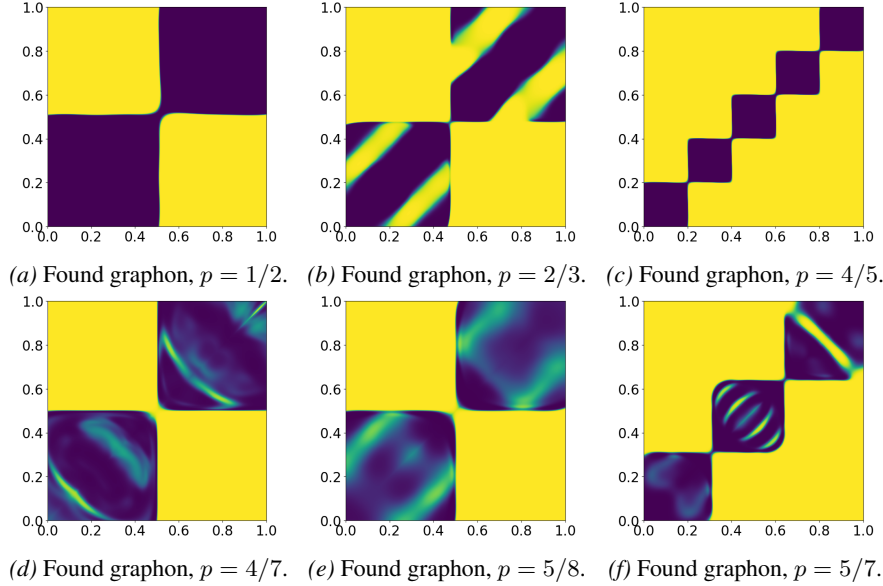


Figure 9. Graphons found for the Petersen graph density minimisation under the edge-density constraint $t(K_2, W) = p$.

E. Additional Experimental Results

In this section, we provide additional experimental results.

We consider the problem of minimising the homomorphism density of the Petersen graph under the edge-density constraint $t(K_2, W) = p$ for various values of p . For minimising the Petersen homomorphism density, there are (to the best of our knowledge) no known results or conjectures, likely due to the graph size and the lack of an obvious extremal structure. We applied our method for $p \in \{1/2, 2/3, 4/5, 4/7, 5/8, 5/7\}$; the learnt graphons are shown in Fig. 9. As there is no known conjectured optimum, we compare our results to those obtained by a hand-crafted graphon motivated from Razborov (2008); see Appendix F for details.

F. Details of Hand-crafted Graphons

In this section, we describe the hand-crafted graphons used as baselines in our experiments. We consider edge densities satisfying $1 - 1/k < p < 1 - 1/(k + 1)$, where k is a positive integer.

The first two instances are graphons arising from the optimal solution to edge-density-constrained K_3 homomorphism-density minimisation. We construct a complete $(k + 1)$ -partite graphon by partitioning $[0, 1]$ into $(k + 1)$ groups, $X_1, \dots, X_{k+1}$, and defining

$$W_i(x_1, x_2) = \begin{cases} 1 & \text{if } x_1 \in X_m, x_2 \in X_n, m \neq n \\ 0 & \text{if } x_1 \in X_m, x_2 \in X_m. \end{cases}$$

We set $|X_1| = \dots = |X_k| = x$ and $|X_{k+1}| = 1 - kx$, and choose x to satisfy the edge-density constraint. This amounts to

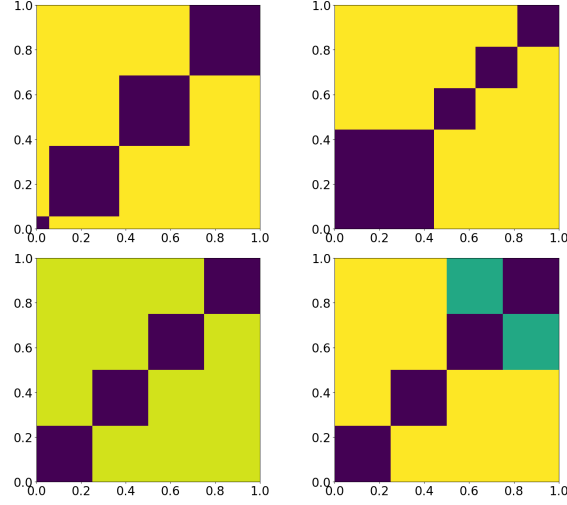


Figure 10. Hand-crafted graphons W_1 (top left), W_2 (top right), W_3 (bottom left), and W_4 (bottom right) with edge density $t(K_2, W) = 0.7$.

solving the quadratic equation

$$k(k-1)x^2 + 2kx(1-kx) = p.$$

This quadratic has two solutions: one with $x > 1 - kx$ (denoted W_1) and one with $x < 1 - kx$ (denoted W_2). The first corresponds to the optimum for K_3 density minimisation.

The second instance is constructed by deleting edges from the complete $(k+1)$ -partite graphon. We partition the interval $[0, 1]$ into k groups $X_1, \dots, X_k, X_{k+1}$ with $|X_1| = \dots = |X_k| = |X_{k+1}| = 1/(k+1)$, and define

$$W_3(x_1, x_2) = \begin{cases} \frac{(k+1)p}{k} & \text{if } x_1 \in X_m, x_2 \in X_n, m \neq n \\ 0 & \text{if } x_1 \in X_m, x_2 \in X_m, \end{cases}$$

$$W_4(x_1, x_2) = \begin{cases} \frac{p(k+1)^2 - k^2 - k + 2}{2} & \text{if } x_1 \in X_1, x_2 \in X_2 \text{ or vice versa} \\ 0 & \text{if } x_1 \in X_m, x_2 \in X_m, \\ 1 & \text{otherwise.} \end{cases}$$

Finally, the last graphon is the constant graphon $W(x_1, x_2) = p$.

We visualised $W_1, \dots, W_4$ for $p = 0.7$ in Fig. 10.